SIXTH LESSON

Probability, Time and the Heat of Black Holes

Along with the major theories that I have already discussed and that describe the elementary constituents of the world, there is another great bastion of physics which is somewhat different from the others. A single question unexpectedly gave rise to it: ‘What is heat?’

Until the mid-nineteenth century, physicists attempted to understand heat by thinking of it as a kind of fluid, called ‘caloric’; or two fluids, one hot and one cold. The idea turned out to be wrong. Eventually James Maxwell and the Austrian physicist Ludwig Boltzmann understood. And what they understood is very beautiful, strange and profound – and takes us into regions which are still largely unexplored.

What they came to understand is that a hot substance is not one which contains caloric fluid. A hot substance is a substance in which atoms move more quickly. Atoms and molecules, small clusters of atoms bound together, are always moving. They run, vibrate, bounce and so on. Cold air is air in which atoms, or rather molecules, move more slowly. Hot air is air in which molecules move more rapidly. Beautifully simple. But it doesn’t end there.

Heat, as we know, always moves from hot things to cold. A cold teaspoon placed in a cup of hot tea also becomes hot. If we don’t dress appropriately on a freezing cold day we quickly lose body heat and become cold. Why does heat go from hot things to cold things, and not vice versa?

It is a crucial question, because it relates to the nature of time. In every case in which heat exchange does not occur, or when the heat exchanged is negligible, we see that the future behaves exactly like the past. For example, for the motion of the planets of the solar system heat is almost irrelevant, and in fact this same motion could equally take place in reverse without any law of physics being infringed. As soon as there is heat, however, the future is different from the past. While there is no friction, for instance, a pendulum can swing forever. If we filmed it and ran the film in reverse we would see movement that is completely possible. But if there is friction then the pendulum heats its supports slightly, loses energy and slows down. Friction